

Michael H. Gross

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RESEARCH INTERESTS

- Autonomous Exploration Systems
- Sensor Fusion and Localization
- Reconfigurable and Multi-Robotic Systems
- Physics-Based Simulation
- Control Systems Design and Optimization
- Dynamic Motion Planning and Trajectory Generation
- Computer Vision and Object Detection
- Digital Twin Development

EDUCATION

Arizona State University, Mesa, AZ

Robotics and Autonomous Systems, *M.S.*

Aug 2024 — May 2025

4+1 Accelerated Program, *with Distinction* — 4.0 GPA

Thesis: Digital Twin Development for Robotic Glovebox Operations

Advisor: Dr. Binil Starly

Robotics Engineering, *B.S.E.*; Minor in Applied Mathematics

Aug 2020 — May 2024

Summa Cum Laude — 4.0 GPA

RESEARCH EXPERIENCE

ASU Robotics and Autonomous Systems Lab, Mesa, AZ

Aug 2024 — July 2025

Graduate Research Assistant

- Developed [armctl](#), a vendor-agnostic Python API controlling industrial robotic arms such as UR, Vention, and Elephant Robotics; achieved around 15 ms latency and removed redundant platform-specific drivers
- Created [mujoco-toolbox](#), a MuJoCo-based digital twin toolkit with XML/URDF merging, inverse kinematics, and trajectory generation; supported reinforcement learning and reproducible closed-loop experiments
- Integrated trajectory optimization, sensor fusion, and adaptive control into cohesive pipelines; demonstrated effectiveness on multi-robot coordination in cluttered and dynamic settings
- Implemented containerized CI/CD workflows that standardized benchmarking, reduced configuration overhead, and increased reliability across research deployments

Robotic Glovebox Digital Twin | Los Alamos National Laboratory

Aug 2024 — Jul 2025

- Built robotic arm digital twin for hazardous glovebox operations with computer vision, motion planning, and control
- Applied deep learning detection with fiducial marker sensor fusion for robust tracking under occlusion/limited visibility
- Optimized path planning for confined workcells, improving accuracy and reliability in high-risk environments

Image Edge Mapping for Object Recognition

Aug — Dec 2022

Evaluated edge detection operators and trained CNNs on 60k+ images for autonomous vehicle detection in MATLAB

Advisor: Dr. Michelle Zandieh

PROFESSIONAL EXPERIENCE

Precision Planting (An AGCO Company), Fargo, ND

May — Aug 2024

Research & Development Intern

- Designed lab-based testing fixtures using Creo & 3D printing to replicate extreme field usage for developmental sensors
- Developed data-cleaning algorithms for optical seeder sensor field data; analysis showed 10% lower detection variance than competitor products, projecting multi-million-dollar savings
- Integrated embedded sensors into the CAN bus with custom drivers, improving compatibility and future development

SELECTED PROJECTS

MuJoCo Toolbox | Published Python Package

Aug 2024 — Present

- Developed comprehensive MuJoCo simulation toolkit streamlining concept-to-simulation workflow for RL, digital twins, and education
- Implemented XML/URDF merging, inverse kinematics, trajectory generation, and automated testing frameworks
- Achieved significant adoption with 1000+ downloads, demonstrating impact in robotics simulation community

Turtlebot4 Predator-Prey Model | Arizona State University

Jan — May 2025

- Built ROS2 nodes for SLAM, computer vision detection, sensor fusion, and localization to enable autonomous navigation
- Designed and tested policy models and motion planning for pursuit-evasion in cluttered environments

- Parkinson's Rehabilitation Device** | Barrow Neurological Institute Jan — May 2025
 - Engineered wearable medical robotics system with IMU and gyroscopic sensors, Bluetooth Low Energy streaming, and interactive GUI
 - Achieved less than 50 ms real-time sensor-to-actuator feedback loop for vibrotactile therapy exercises
- Frog Gait Reproduction** | Arizona State University Aug — Dec 2024
 - Developed foldable bio-inspired robotic frog using five-bar linkage mechanism with material parameter optimization
 - Applied non-linear optimization using MuJoCo simulation tools for locomotion performance enhancement
- Tilt-Up Window Automation** | Summit Automation Aug 2023 — May 2024
 - Developed 400 lb IoT device with linear actuators, embedded control, and custom PCB; delivered commercially
 - Integrated voice recognition, embedded design, and remote monitoring capabilities
- Smart Umbrella for Wheelchairs** | Arizona State University Jan — May 2023
 - Designed motorized canopy with real-time climate sensor control and 2DOF control system
 - Implemented custom PCB design and integrated smart sensors for autonomous weather response

LEADERSHIP AND ACTIVITIES

- FIRST Tech Challenge (FTC) – Pioneer Robotics Club**, Tucson, AZ Aug 2022 — Present
 - Head Programming Mentor, Original Founding Member*
 - Taught sensor fusion, Kalman filtering, world-centric localization, multithreading, and path planning for automation
 - Mentored student teams in advanced robotics concepts including computer vision and autonomous navigation
- Design & Manufacturing Challenge (MR+A)** Oct 2024 — Present
 - Digital manufacturing innovation competition with focus on advanced manufacturing techniques
- FLL Mentor/Volunteer** Aug 2015 — Present
 - Long-term commitment to STEM mentorship and event volunteering in robotics education
- ASU Esports: Team Manager, Social Media Lead** Aug 2020 — Present
 - Leadership role in competitive gaming organization with team coordination and marketing responsibilities

HONORS AND AWARDS

- 1st Place – Robotics & Art Competition** AY 2024—25
 - ASU Ira A. Fulton Schools of Engineering recognition for innovative robotics design
- Graduate Scholarship** AY 2024—25
 - ASU Ira A. Fulton Schools of Engineering merit-based funding for graduate studies
- Dean's List** Fall 2020 — Spring 2024
 - ASU Ira A. Fulton School of Engineering academic excellence recognition

TECHNICAL SKILLS

- Languages & Frameworks:** Python, C/C++, Rust, MATLAB, ROS2, OpenCV, MuJoCo, Gazebo
- Tools & Systems:** Git, Docker, CI/CD, Linux (Ubuntu, Arch), Embedded Systems, SPI, I2C, UART
- CAD & Fabrication:** SolidWorks, Fusion 360, KiCad, Cadence, 3D Printing, CNC Machining, Laser Cutting
- Robotics Hardware:** UR Cobots, myCobot, Vention MachineMotion, Elephant Robotics

CERTIFICATIONS

- UR Cobot eSeries (2023):** Core Track, Pro Track — Industrial robot programming and safety
- MATLAB (2022):** Control Design, Deep Learning, Machine Learning, Simscape, Image/Signal Processing

PUBLICATIONS & MEDIA

- Software Packages:**
 - [mujoco-toolbox](#) — MuJoCo simulation toolkit with 1000+ downloads
 - [armctl](#) — Unified robotic arm control API for multiple OEMs
 - [spikeble](#) — Bluetooth Low Energy interface for robotics applications
- Media Coverage:**
 - [ASU Dream Competition Winner Coverage, 2025](#)
 - [AZFLL State Competition Coverage, 2014](#)

SUMMARY

Robotics researcher and software engineer with expertise in perception, planning, control, and embedded systems. Experienced across the full mechatronics stack, from low-level firmware and hardware interfaces to high-level libraries and frameworks. Dedicated to delivering modular and scalable solutions that advance robotics in research, product development, and deployment. Strong record of open-source contribution and practical application of research findings.